

Instructions to the Students:

1. Question No 1 is compulsory and carry 10 marks whereas Question No 2 to Question 7 carries 12 marks each.
2. Attempt any five Questions from Question No 2 to Question No 7.
3. Illustrate your answers with neat sketches, diagrams etc. wherever necessary.
4. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is a part of examination.
5. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Select an appropriate option for each of the following:

(10)

The function of the signal manipulation element in a generalized measurement system is to.....

- a) change the quantity under measurement to an analogous signal
- b) change the magnitude of the input signal while retaining its identity.
- c) perform nonlinear operations like filtering, chopping, clipping and clamping.
- d) none of the above

Middle value of a set of an "odd" number of readings, if variables are arranged in numerical order is known as

- a) arithmetic mean
- b) median
- c) mode
- d) deviation

The sum of all deviations is

- a) always zero
- b) never zero
- c) always positive
- d) none of these

The source of emission of electrons in CRT is.....

- a) PN junction diode.
- b) heated cathode
- c) control grid
- d) all of these

5. The input resistance of cathode ray oscilloscope is of the order of

- a) tens of ohm
- b) megaohm
- c) kiloohm
- d) fraction of an ohm

- Q. 2. Attempt the following (12)
- a) Explain speed of response, fidelity, lag and dynamic error.
 - b) Draw the block diagram and explain the working of vector impedance meter.
- Q. 3. Attempt the following (12)
- a) Explain arithmetic mean, deviation, average deviation, standard deviation and variance in brief.
 - b) With block diagrams, explain how digital frequency meter can be used to measure time interval and frequency ratio.
- Q. 4. Attempt the following (12)
- a) Write a note on cathode ray tube.
 - b) Explain in brief, various types of oscilloscope probes.
- Q. 5. Attempt the following (12)
- a) Draw the block diagram and explain the working of a heterodyne type wave analyzer.
 - b) Draw the block diagram of basic spectrum analyzer using swept receiver design and explain its operation.
- Q. 6. Attempt the following (12)
- a) Write a note on signal separation device used in network analyzer.
 - b) Explain the most commonly used receiver techniques in network analyzer.
- Q. 7. Attempt the following (12)
- a) Explain computer controlled testing of an audio amplifier using block diagram.
 - b) Write a note on IEEE 488 standard.

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